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Section : BAI-2A

Labtask : 13

Question 1:

Code:

```
#include<iostream>

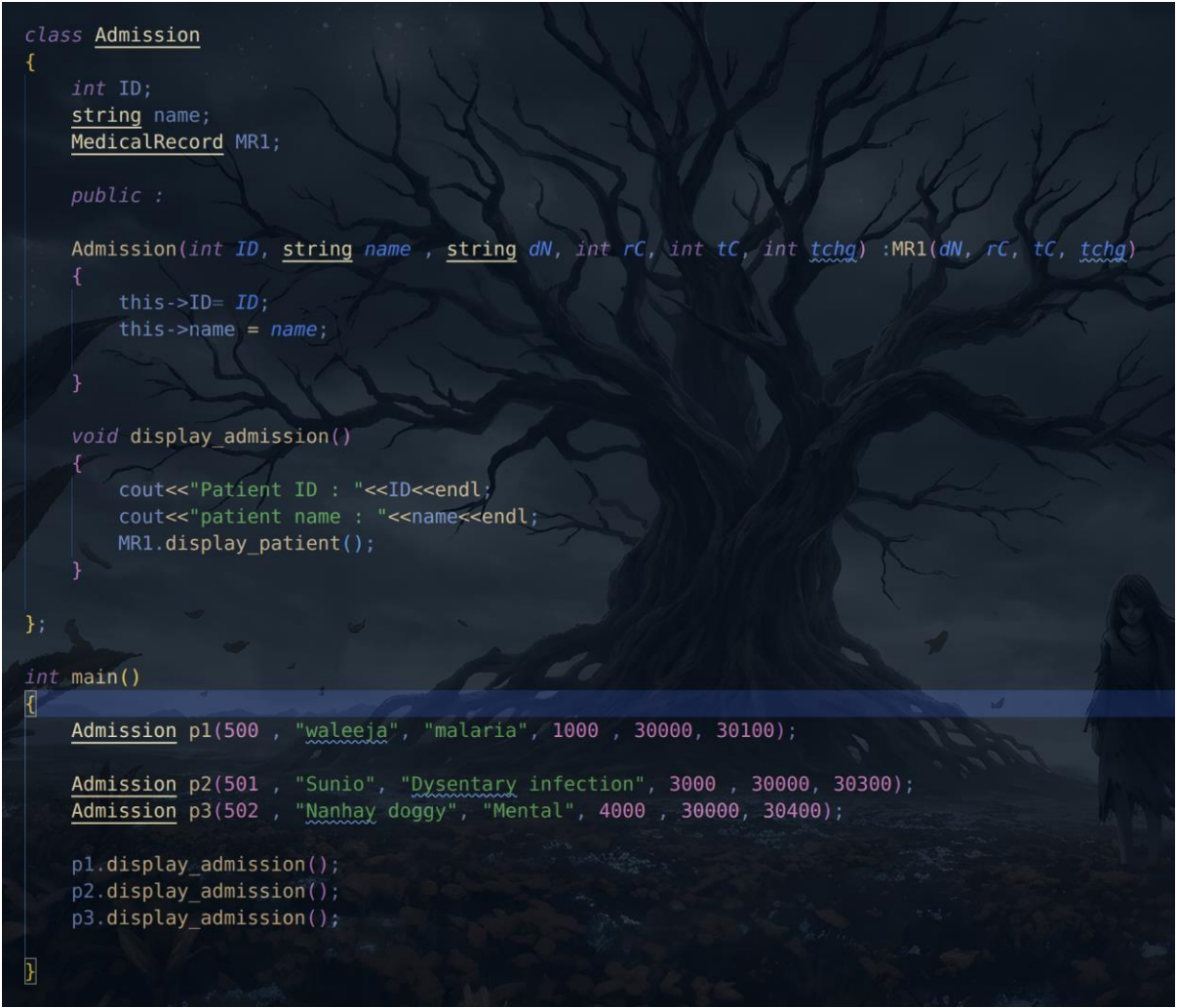
using namespace std;

class MedicalRecord
{
    string diseaseName;
    int roomCharges;
    int treatmentCost;
    int totalCharges;

    public :

    MedicalRecord(string dN, int rC, int tC, int tchg)
    {
        diseaseName = dN;
        roomCharges = rC;
        treatmentCost = tC;
        totalCharges = tchg;
    }

    void display_patient()
    {
        cout<<"Disease name : "<<diseaseName<<endl;
        cout<<"Room Charges : "<<roomCharges<<endl;
        cout<<"Treatment Cost : "<<treatmentCost<<endl;
        cout<<"Total Charges : "<<totalCharges<<endl;
        cout<<endl;
    }
};
```



```
class Admission
{
    int ID;
    string name;
    MedicalRecord MR1;

    public :

    Admission(int ID, string name , string dN, int rC, int tC, int tchg) :MR1(dN, rC, tC, tchg)
    {
        this->ID= ID;
        this->name = name;
    }

    void display_admission()
    {
        cout<<"Patient ID : "<<ID<<endl;
        cout<<"patient name : "<<name<<endl;
        MR1.display_patient();
    }
};

int main()
{
    Admission p1(500 , "waleeja", "malaria", 1000 , 30000, 30100);

    Admission p2(501 , "Sunio", "Dysentary infection", 3000 , 30000, 30300);
    Admission p3(502 , "Nanhay doggy", "Mental", 4000 , 30000, 30400);

    p1.display_admission();
    p2.display_admission();
    p3.display_admission();
}
```

Output:

Patient ID : 500
patient name : waleeja
Disease name : malaria
Room Charges : 1000
Treatment Cost : 30000
Total Charges : 30100

Patient ID : 501
patient name : Sunio
Disease name : Dysentary infection
Room Charges : 3000
Treatment Cost : 30000
Total Charges : 30300

Patient ID : 502
patient name : Nanhay doggy
Disease name : Mental
Room Charges : 4000
Treatment Cost : 30000
Total Charges : 30400

Question 2:

Code:

```

#include<iostream>
using namespace std;

class Professor
{
    int ID;
    string name;
    string specialization;

    public :
    Professor(int id, string n, string s) : ID(id), name(n), specialization(s) {};

    void display_professor()
    {
        cout<<"Professor ID : "<<ID<<endl;
        cout<<"Professor Name "<<name<<endl;
        cout<<"specialization : "<<specialization<<endl;
    }
};

```

```

21 class Department
22 {
23     string depName;
24     Professor * prof;
25
26     public :
27     Department(string dN, Professor* p)
28     {
29         depName = dN;
30         prof = p;
31     }
32
33     void display_dep()
34     {
35         cout<<"Department name : "<<depName<<endl;
36         prof->display_professor();
37         cout<<endl;
38     }
39 };
40
41 int main()
42 {
43     Professor p1(1, "Awais" , "ENT");
44     Professor p2(2, "Talha", "orthopadic surgen" );
45
46
47     Department D1("ENT department ", &p1);
48     Department D2("orthopadic department" , &p2);
49
50     D1.display_dep();
51     D2.display_dep();
52
53 }

```


Output:

```
Department name : ENT department
```

```
Professor ID : 1
```

```
Professor Name Awais
```

```
specialization : ENT
```

```
Department name : orthopadic department
```

```
Professor ID : 2
```

```
Professor Name Talha
```

```
specialization : orthopadic surgen
```